

LAB EXPERIMENT 6
EMOTION RECOGNITION USING LANDMARK FEATURES
AND DEEP NEURAL NETWORKS

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Acknowledgment: I acknowledge all of the work (including figures and codes) belongs to me and/or persons who are referenced.

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LAB 6 SUMMARY

In this lab, we explored a complete machine learning pipeline for recognizing human emotions using facial landmark features and a deep neural network (DNN). The goal was to understand how facial feature data can be processed and used effectively for emotion classification, with a strong focus on both the image processing and deep learning aspects.

We started by working with Dlib, a library that uses Histogram of Oriented Gradients (HOG) and regression trees to detect faces and extract 68 key facial landmarks from each image. These landmarks represent key regions like the jawline, eyebrows, nose, eyes, and mouth. Rather than using full image data, we converted these landmarks into a flat vector format:

$[(x_1, y_1), (x_2, y_2), \dots, (x_{68}, y_{68})] \rightarrow [x_1, y_1, x_2, y_2, \dots, x_{68}, y_{68}]$

This preprocessing simplifies the model input and allows us to train a more efficient neural network.

To make the data more consistent and suitable for training, we normalized the coordinates. This involved:

- Aligning the nose tip (point 34) to the origin (0, 0),
- Scaling the width using point 1 (right jaw),
- Scaling the height using point 9 (top of nose bridge).

The final data is saved in CSV format, with the last column representing the emotion label (from 0 to 7), covering emotions such as angry, happy, sad, neutral, and more.

Next, we built a deep neural network using TensorFlow. The model consisted of fully connected (dense) layers, with 136 input nodes (68 landmarks $\times$ 2) and an output layer for the 8 emotion classes. We used ReLU activations and trained the model on the landmark dataset we created. This step showed how simplifying input features can lead to faster training and decent accuracy without needing a complex CNN architecture.

We tested the model and recorded the accuracy results. The network was able to learn meaningful patterns in the landmark vectors and classify emotions accordingly.

Key Takeaways:

- Facial landmarks are a powerful and compact representation of facial expressions.
- Preprocessing through normalization improves training stability and performance.
- A simple DNN can perform well when the input features are well-structured.

- Building and understanding the full pipeline (from raw image to emotion prediction) gave me deeper insight into real-world AI applications.

Tools Used:

- Dlib for face detection and landmark extraction
- OpenCV for image processing
- TensorFlow for neural network construction
- Anaconda & Jupyter Notebook for environment setup and execution

This lab helped me understand how a well-designed preprocessing strategy can reduce complexity and still deliver strong results. More importantly, it gave me a clear view of how traditional computer vision techniques can be combined with modern deep learning methods to create a functional and efficient emotion recognition system.